



INTERIM PRESENTATION

Name

Date



Research Question

- Re-state the research question or goal of the project
- Re-phrase the research question into a statistically testable hypothesis or series of hypotheses
 - *Which variables are your outcomes?*
 - *Primary explanatory variables?*
 - *Covariates, precision variables, confounders?*
- If something about the research question is unclear to you or if you are unsure if your statistical hypotheses address the investigator's research question, ask the investigator in your presentation

Data Overview and Descriptive Statistics

- Basic descriptive statistics of the key variables in the dataset
 - *Number of observations*
 - *Number missing*
 - *Continuous variables: mean and SD (or median, IQR), min and max*
 - *Categorical: N in each category, percent or proportion*
 - *Consider reporting descriptive statistics by group if your PEV is a categorical variable*
- Are there any problems with the data that need to be resolved or considered before you begin your analysis?
 - *Outliers*
 - *Missing data*
 - *Variables that need transformation to meet model assumptions*
 - *Categorical variables with very few observations in certain categories*
- If space permits, you might include a plot on this slide to illustrate important aspects of your initial data exploration

Analysis Plan

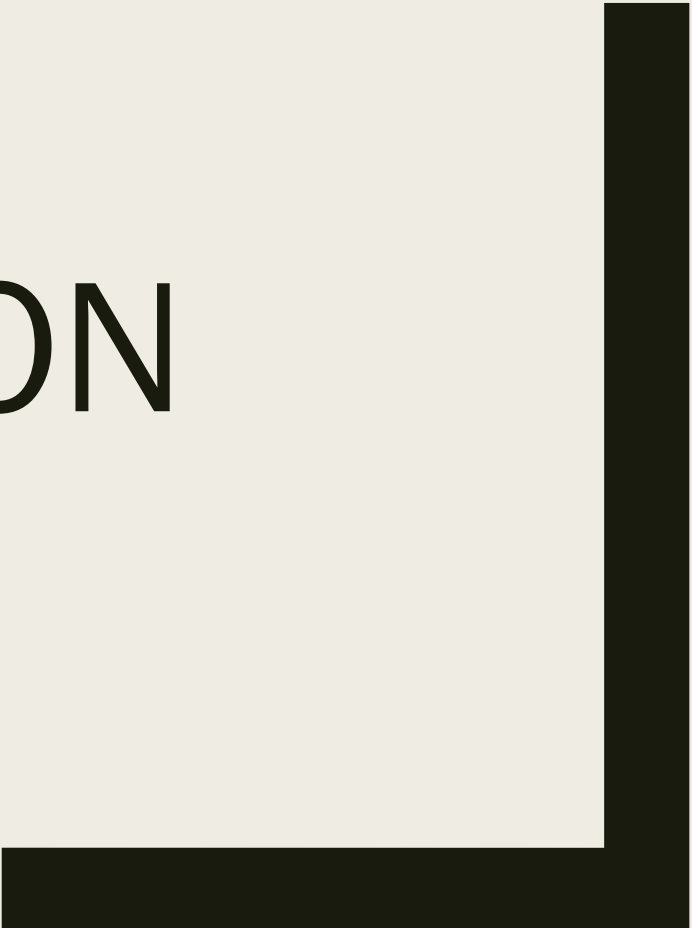
- How do you plan to model the data?
 - *Linear regression, logistic regression, etc.?*
- What variables will be included in your model?
 - *Outcome, PEV*
 - *Confounders, Mediators, Precision Variables, Interaction Terms, if applicable*
 - *If you are unsure of which variables will be in your final model a priori, describe your model selection strategy*
- What statistical tests will you perform to address the research question?
 - *Will you test specific regression coefficients or combinations of regression coefficients?*
 - What would a significant result from these tests indicate?
 - *What significance threshold will you use?*
 - Do you need to worry about multiple comparisons?



FINAL PRESENTATION

Name

Date



Research Question

- State the research question and the statistical hypotheses tested

Data Overview

- Summarize the data used in the analysis
 - *How many subjects were in the initial dataset you received from the investigator?*
 - *How many were included in the final analysis?*
 - If these numbers are not the same, why were some subjects excluded?
 - *Did you make any transformations or other changes to the data, such as collapsing groups in categorical variables?*
 - If so, why?
 - *Descriptive statistics for the final analysis dataset*

Analysis

- Brief explanation of the statistical analyses
 - *Type of model (ex. Linear regression, logistic regression, etc.)*
 - *Variables included in the models and their roles*
 - *Statistical hypotheses tested*

Results

- Summarize the results as they relate to the statistical hypotheses tested
 - *You do not need to interpret or discuss every parameter in your regression model*
 - *Interpret the regression parameters and combinations of parameters that address the research question*
 - *Be sure to include p-values, point estimates, and 95% confidence intervals.*
 - Ex. Controlling for age, males had significantly higher blood glucose than females ($p=0.001$). On average males' blood glucose levels were 2.5 units higher than womens' (95% CI: 1.5 to 3.5).
 - Ex. We did not find a significant effect of smoking ($p=0.13$). The average difference in BMI between non-smokers and smokers was 0.3 kg/m² (95% CI: -0.3, 0.6, with negative values indicating higher BMI in smokers).

Results, continued

Include plots to help illustrate your results if appropriate

Discussion

- Summary of how the results answer the research question
- Limitations of the analysis